Advanced Statistical Tools for Data Science

Course Placement: It's an elective for B Tech ICT, B Tech Maths and Computing, Masters in Data Science, M Tech, Ph.D.

Prerequisite: A basic course in probability & statistics

Course format (3-1-0-4): Three hours' lecture and one-hour tutorial every week.

Course Content:

Modelling data using regression*:* Regression problem, Regression models: parametric & non-parametric, Model selection, Model estimation & Inference, Multilevel/Hierarchical regression models

Resampling Techniques- Jackknife & Bootstrap: Estimation of Bias and Variance, Resampling based inference (estimation, testing and prediction)

Bayesian Data Analysis: Frequentist versus Bayesian Inference, Prior & posterior distributions, Credible intervals, Bayes' factor

Missing Data Analysis: Different missingness mechanisms, Different Imputation methods, EM Algorithm,

Causal Inference: Causation versus correlation, Potential outcome framework for causal inference, Causal inference using data from (i) randomized experiments, (ii) observational studies, (iii) natural experiments.

References:

1. Data analysis using regression and multilevel hierarchical models – Andrew Gelman and Jennifer Hill (Cambridge university press)
2. An introduction to the bootstrap — Bradley Efron and Robert Tibshirani (CRC Press)
3. The jackknife, the bootstrap and other resampling plans—Bradley Efron (CBMS-NSF Regional Conference Series)
4. Bayesian data analysis—Andrew Gelman, John B. Carlin, Hal H. Stern, David B. Dunson, Aki Vehtari, Donald B, Rubin (Chapman Hall/CRC Press)
5. Statistical Analysis with Missing Data: Roderick J. A. Little and Donald Rubin (Wiley Interscience)
6. Multiple imputation for non-response in surveys: Donald Rubin (John Wiley)
7. The book of why: Judea Pearl and Dana Mackenzie (Basic Books)
8. Observation and experiment: Paul R. Rosenbaum (Harvard University Press)
9. Counterfactuals and causal inference: Stephen L Morgan and Christopher Winship (Cambridge University Press)
10. Causal Inference for statistics, social and biomedical sciences: Guido W. Imbens and Donald B. Rubin (Cambridge University Press)

Assessment method: Quizzes, Assignments and an end semester examination

Lecture Schedule

Sl. No.	Description	No. of Lectures
	Modelling Data Using Regression	
1.	**General regression problem**	**2**
	The general model, Explanation or Prediction, Regression function, Trade-off between predictive accuracy and explanation	
2.	**Linear regression**	**5**
	The model & assumptions, estimation, inference, measures of model fit, detection of outliers and leverage points, handling heteroscedasticity	
3.	**Logistic regression**	**2**
	The model & assumptions, interpretation of coefficients, estimation, inference, evaluating, checking and comparing fitted logistic regressions, Identifiability and separation	
4.	**Generalized linear models**	**3**
	Poisson regression, logistic-binomial model, probit regression, ordered and unordered categorical regression, robust regression using t-model	
5.	**Multilevel regression**	**3**
	Varying-intercept and varying-slope models, clustered data, repeated measurements, time-series cross sections, and other non-nested structures, indicator variables and fixed or random effects	
	Resampling Techniques	

1.	**Jackknife**	**2**
	The jackknife estimate of bias and variance of sample median, sample trimmed mean, ratio estimate and regression coefficients	
2.	**Bootstrap**	**5**
	Parametric bootstrap, Smoothed bootstrap, bootstrap methods for more general problem, bootstrap estimate of bias, non-parametric bootstrap, bootstrap confidence intervals, bootstrap P-values	
	Bayesian Data Analysis	
1.	Subjective and frequentist probability, paradoxes of classical statistical inference, Elements of Bayesian inference, Priors, posteriors	**4**
2.	Common problems of Bayesian inference: Point estimates, testing using Bayes' factor, credible intervals, testing a sharp null hypothesis using credible intervals, prediction of a future observation,	**6**
	Missing Data Analysis	
1.	Different missingness mechanisms, Traditional methods for dealing with missing data: list wise deletion, pairwise deletion, single imputation methods, regression imputation	2
2.	Maximum likelihood method for handling missing data, EM algorithm,	4
3.	Bayesian estimation and multiple imputation, Illustration through examples	2
	Causal Inference	
1.	Counterfactuals and the potential outcome model, the average treatment effect, the stable unit treatment value assumption	2
2.	Estimating average treatment effect by random assignment, from observational studies and natural experiments	3

Course Outcomes: This course comprises five advanced statistical topics that are often used by data scientists for analyzing data. After completion of this course, students are expected to

- Learn the core concepts underpinning the theoretical and methodological aspects of each of these topics
- Analyse data arising in different disciplines to answer interesting research questions
- Learn to implement these methodologies for analyzing data using R/Python
- Get a feel for data driven decision making

P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
X			X								